Swiss Paraplegic Research

# Reproducible Reports – Quarto

## Workshop R-Introduction

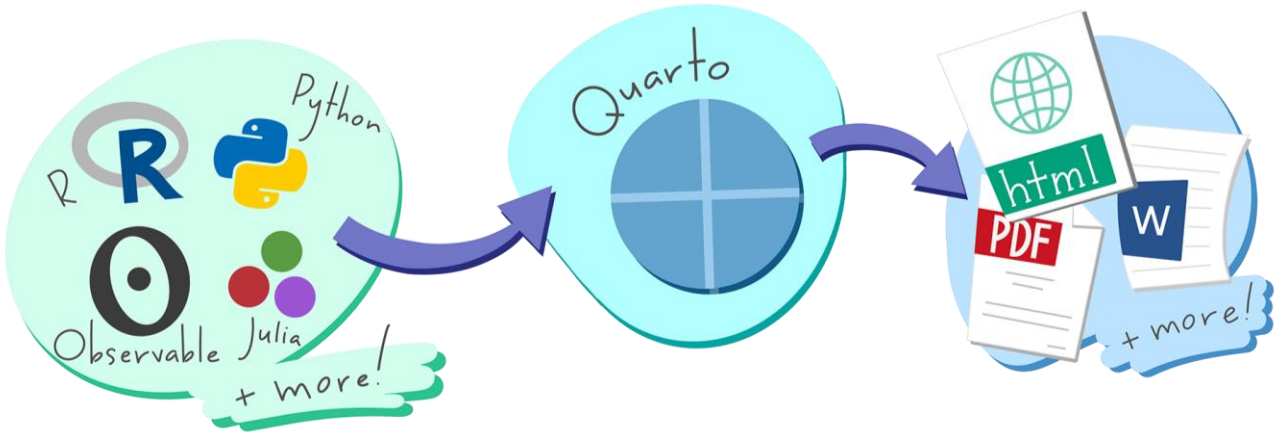
Andreas Limacher, Group Leader

19.10.2026



# Quarto

## Overview



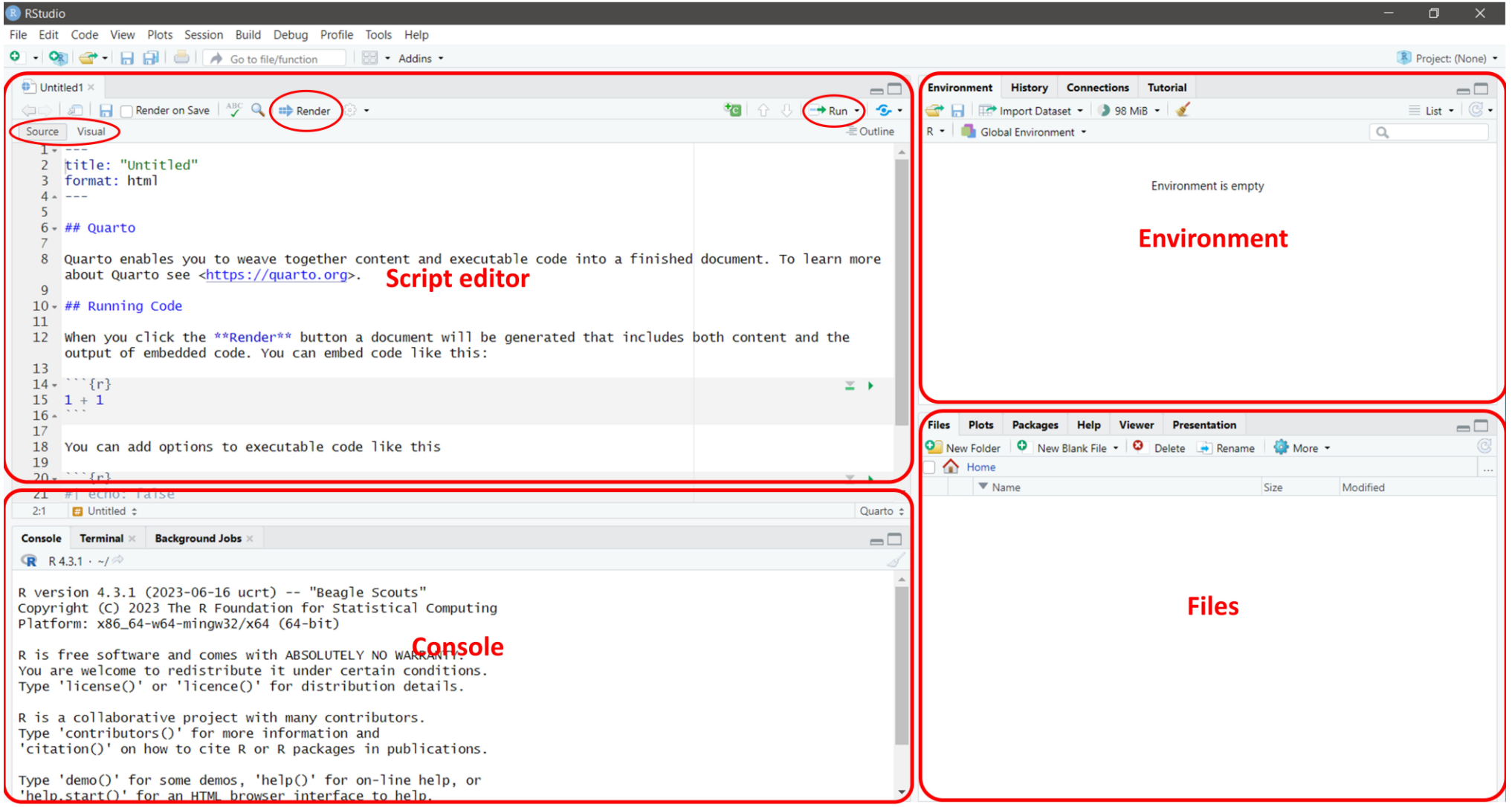

**What is Quarto?**

- Open-source publishing system
- Further development of R-Markdown
- Combines text, R code, and output in a single document
- Publishes reproducible output in HTML, PDF, and MS Word

**What is it used for?**

- 📊 Reports & statistical analyses — automatically generate tables and figures
- 📄 Manuscripts & supplementary materials — keep analysis and reporting together
- 🎤 Presentations & slides — create slides directly from your analysis
- 🌐 Websites, blogs & teaching materials — publish the same content in different formats

# R Studio


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Project: (None)
Untitled1
Render on Save
Render
Run
Source
Visual
Outline
1 ---
2 title: "Untitled"
3 format: html
4 ---
5
6 ## Quarto
7
8 Quarto enables you to weave together content and executable code into a finished document. To learn more about Quarto see <https://quarto.org>.
9
10 ## Running Code
11
12 When you click the **Render** button a document will be generated that includes both content and the output of embedded code. You can embed code like this:
13
14 ```{r}
15 1 + 1
16 ```
17
18 You can add options to executable code like this
19
20 ```{r}
21 #| echo: false
Script editor
2:1
Untitled
Quarto
Console
Terminal
Background Jobs
R 4.3.1 · ~/
R version 4.3.1 (2023-06-16 ucrt) -- "Beagle Scouts"
Copyright (C) 2023 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64 (64-bit)
R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Console
Environment
History
Connections
Tutorial
Import Dataset
98 MiB
List
R
Global Environment
Environment is empty
Environment
Files
Plots
Packages
Help
Viewer
Presentation
New Folder
New Blank File
Delete
Rename
More
Home
Name
Size
Modified
Files

# Quarto

## Overview




**What does a Quarto script file contain?**

- YAML header at the beginning to define metadata and document-wide settings
- Markdown text for structuring and describing
- Code chunks to execute R code

**YAML header**

```
---
title: "Quarto"
format: html
---
```

# YAML header

## R-Markdown vs Quarto

| R Markdown | Quarto |
|---|---|
| .Rmd file | .qmd file |
| output: | format: |
| html_document: | html: |
| pdf_document: | pdf: |
| code_folding: | code-fold: |
| R Markdown-specific options | More general, language-independent options |
| Primarily associated with R | R, Python, Julia, Observable, etc. |

# Markdown text and formatting




- Headings:

```
# Heading 1
## Heading 2
### Heading 3
```

- Bold and italic text:

```
**bold**
*italic*
```

- Listings:
  - bullet points

```
- Bullet point
- Another bullet point
```

  - numbered lists

```
1. First item
2. Second item
```

# R code chunks

## R-Markdown vs Quarto

- R-Markdown (.Rmd)

````
```{r, echo=FALSE, warning=FALSE, fig.width=7}

plot(data$x, data$y)
```
````

- Quarto (.qmd)

````
```{r}
#| echo: false
#| warning: false
#| fig-width: 7

plot(data$x, data$y)
```
````

R Markdown and Quarto use essentially the same code-chunk concept.

The big change is that Quarto moves R chunk options from {r, ...} into #| lines.

# Inline R code

- Calculated values can be integrated into text
- Avoids mistakes when updating the data and/or the document (no manual transfer)
- Inline code, beginning with a backstick and r:

````
The mean is`r mean(c(1, 2, 3, 4, 5))`.
````

- Combination of code chunk and inline code; calculate an object in a chunk and use it in the text:

````
```{r}
model <- lm(cty ~ hwy, data = mpg)
b <- coef(model)
p <- summary(model)$coefficients[2,4]
```

The model coefficient is b = `r round(b, 2)` (p = `r round(p, 3)`).
````

# Example

```yaml
---
title: "Hello, Quarto"
format: html
editor: visual
---
```

```r
{r}
#| label: load-packages
#| include: false

library(tidyverse)
library(palmerpenguins)
```

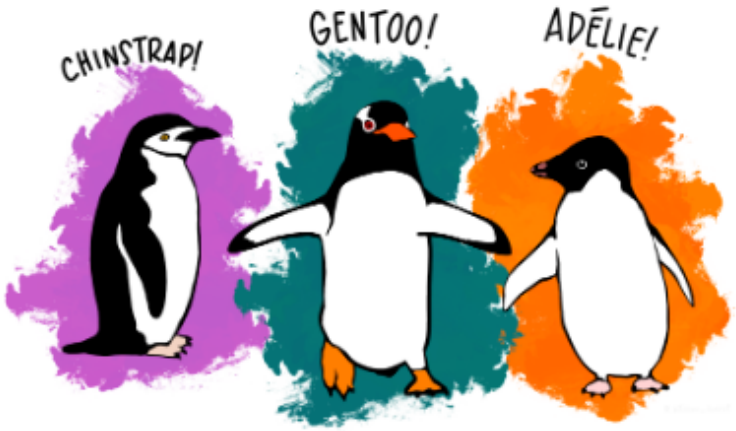
## Meet Quarto

Quarto enables you to weave together content and executable code into a finished document. To learn more about Quarto see https://quarto.org.

## Meet the penguins

The `penguins` data from the **palmerpenguins** package contains size measurements for `{r} nrow(penguins)` penguins from three species observed on three islands in the Palmer Archipelago, Antarctica.

The plot below shows the relationship between flipper and bill lengths of these penguins.




Source: https://quarto.org/

# Display options for output

- By default, the code and its output are displayed within the rendered document.
- Display options in single R chunks override overall display options given in header

| | R code (single chunks) | YAML header (allover) |
|---|---|---|
| Hide warnings | #\| warning: false | execute:<br>  warning: false |
| Hide code | #\| echo: false | execute:<br>  echo: false |
| Hide results | #\| output: false | execute:<br>  output: false |
| Hide code and results | #\| include: false | execute:<br>  include: false |
| Fold/collapse code | #\| code-fold: true | format:<br>  html:<br>  code-fold: true |

# Plot appearance

- Specify label, caption, and size of plot:

````
```{r}
#| label: fig-scatterplot
#| fig-cap: "City and highway mileage."
#| fig-width: 6
#| fig-height: 3.5

ggplot(mpg, aes(x = hwy, y = cty, color = cyl))
```
````

- Reference to figure in text using the label:

```
@fig-scatterplot shows a positive, strong, and linear relationship between
the city and highway mileage of these cars.
```

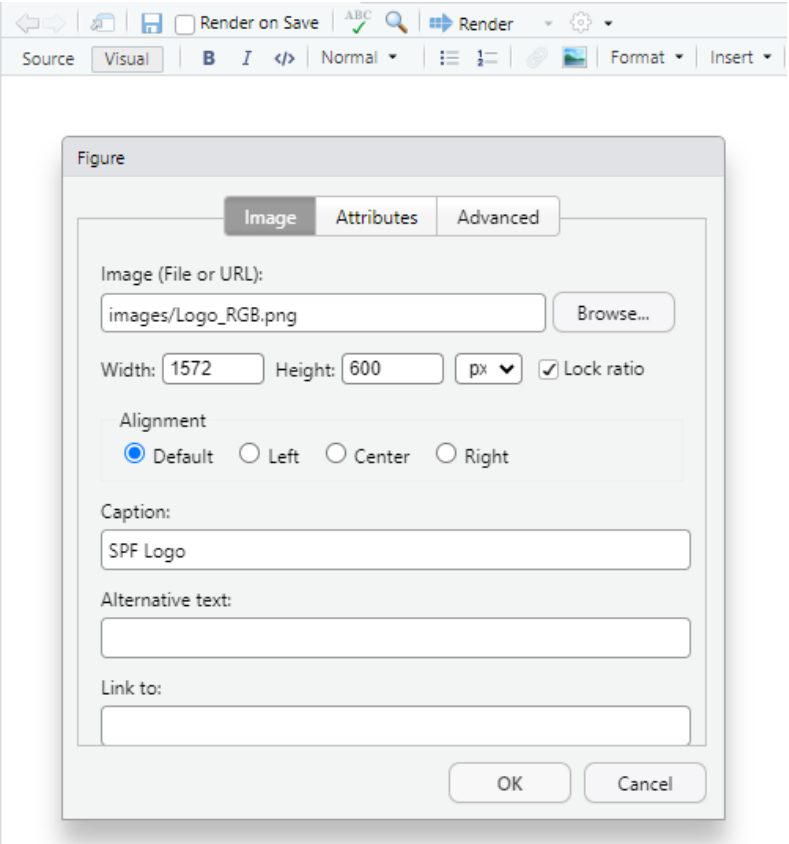
# External image

- Embed an image from an external file:

  `> Visual editor -> Insert menu -> Browse to the image and add a caption`

- The caption and file name will appear in the text body:

  `![SPF Logo](images/Logo_RGB.png)`



# Adding tables

- Specify caption and formatting of the table using the knitr- or gt-package:

````
```{r}
#| label: tbl-data
#| tbl-cap: "First five records of fuel-economy data"

library(knitr)
library(gt)

mpg[1:5, ]
kable(mpg[1:5, ], )
gt(mpg[1:5, ], )
```
````

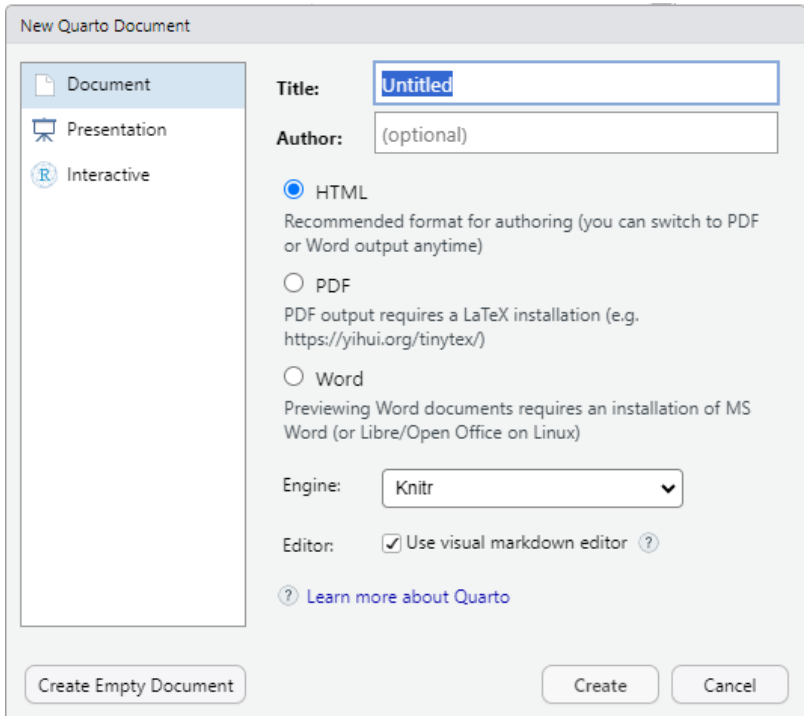
```
-> default formatting
-> kable formatting
-> gt formatting
```

# Output formats

- Formats:
  - HTML (default)
  - PDF (LaTeX installation or TinyTeX R-package required)
  - Word, PowerPoint, etc.
- Multiple formats possible, specified in YAML header, e.g.

```
format:
  - html
  - pdf
  - docx
```

- Via menu:
  - File > New File > Quarto Document…



# Table of content and references

- Table of content and section numbering, specified in YAML header

```
toc: true
number-sections: true
```

- References

YAML header: `bibliography: references.bib`
In-text: `[@knuth1984]`



# Exercise




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